

Data Analytics II: Empirical Research Methods

– Assignment 1 –

Federica Mascolo

Valentina Sontheim

Lea Tschan

Data

The dataset **Assignment1** contains survey data from 1977-1978 on doctor visits in Johto. The variables are:

- visits - Average number of days a person visited at least one doctor per week in the past 2 weeks.
- gender - Character indicating gender.
- age_squared - Age in years squared.
- income - Annual income in tens of thousands of dollars.
- illness - Number of illnesses in past 2 weeks.
- reduced - Number of days of reduced activity in past 2 weeks due to illness or injury.
- health - General health questionnaire score using Goldberg's method.
- private - Does the individual have private health insurance?
- freepoor - Does the individual have free government health insurance due to low income?
- freerepat - Does the individual have free government health insurance due to old age, disability or veteran status?
- nchronic - Is there a chronic condition not limiting activity?
- lchronic - Is there a chronic condition limiting activity?

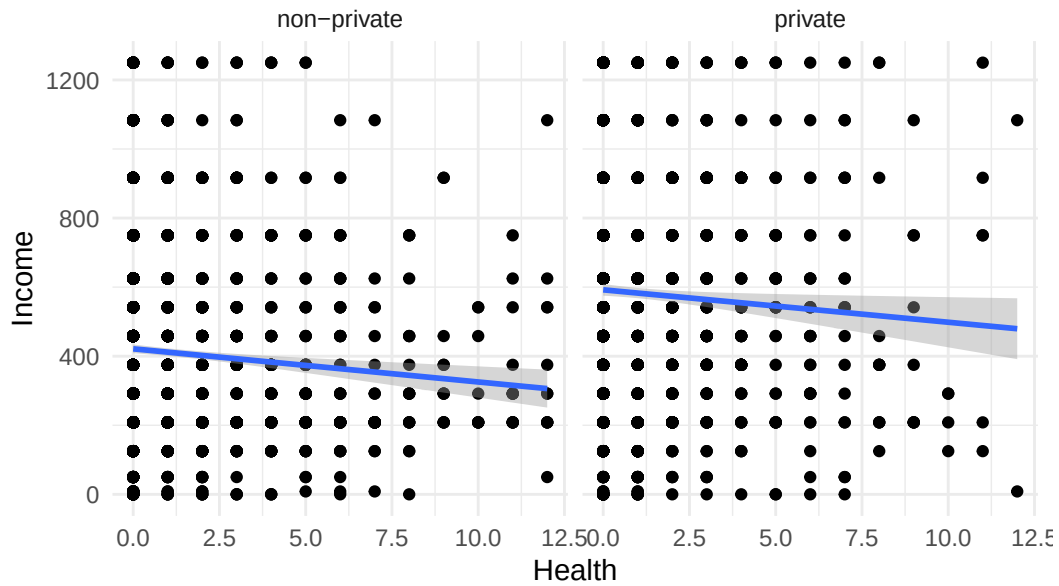
Exercise 1

- Download the data from Canvas and import the dataset in R. Clean the dataset (e.g. checking for extreme values and such that the variables make sense). You need to use the command that we saw in the first exercise session. [0.5 point]
- Compute the monthly income in dollars and its average and variance (be careful to use the right unit). Do you think that the monthly income in dollars correlates with **health**? Write what you would expect ex ante and what you observe in the data. Write no more than 3 sentences. [1.5 points]

Exercise 2

- Generate two datasets named 'data_private' and 'data_age' with the required variables to produce a plot to show the correlation between **health** and **income** for people in different insurance types and to observe the correlation between **health** and **income** separately for individuals above and below median age. [0.5 point]
- Add a column to the dataset 'data_age' with a factor variable containing two age classes: (i) if the individual is above the median age in the sample; (ii) if the individual is below the median age in the sample. (Hint: use the **ifelse()** and **factor()** functions) [0.5 point]
- Write the code to generate the following graphs. What can you conclude from the graphs with respect to association vs. causation or across different conditioning variables (insurance types and age)? (max 200 words) [2 points]

Correlation between Health and Income by Insurance Type



Correlation between Health and Income by Age Class

